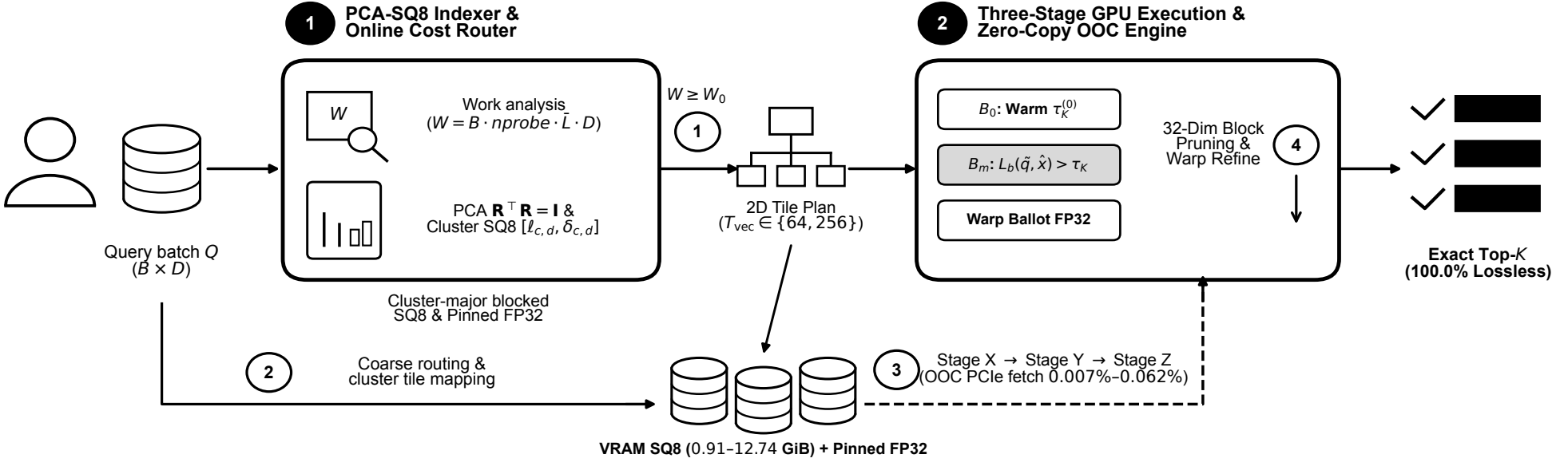

1
PCA-SQ8 Indexer &
Online Cost Router
2
Three-Stage GPU Execution &
Zero-Copy OOC Engine
W
Work analysis
($W = B \cdot nprobe \cdot \bar{L} \cdot D$)
PCA $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ &
Cluster SQ8 [$\ell_{c,d}$, $\delta_{c,d}$]
Cluster-major blocked
SQ8 & Pinned FP32
$W \geq W_0$
1
2D Tile Plan
($T_{\text{vec}} \in \{64, 256\}$)
B_0: Warm $\tau_K^{(0)}$
B_m: $L_b(\tilde{q}, \hat{x}) > \tau_K$
Warp Ballot FP32
32-Dim Block
Pruning &
Warp Refine
4
Exact Top-K
(100.0% Lossless)
Query batch Q
($B \times D$)
2
Coarse routing &
cluster tile mapping
3
Stage X → Stage Y → Stage Z
(OOC PCIe fetch 0.007%–0.062%)
VRAM SQ8 (0.91–12.74 GiB) + Pinned FP32